

A Comparison of Lake Trout Spawning, Fry Emergence, and Habitat Use in Lakes Michigan, Huron, and Champlain

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ABSTRACT. Restoration of self-sustaining populations of lake trout is underway in all of the Great Lakes and Lake Champlain, but restoration has only been achieved in Lake Superior and in Parry Sound, Lake Huron. We evaluated progress toward restoration by comparing spawning habitat availability, spawner abundance, egg and fry density, and egg survival in Parry Sound in Lake Huron, in Lake Michigan, and in Lake Champlain in 2000–2003. Divers surveyed and assessed abundance of spawners at 5 to 15 sites in each lake. Spawning adults were sampled using standardized gill nets, eggs were sampled using egg bags, and fry were sampled using emergent fry traps and egg bags left on spawning reefs overwinter. Spawning habitat was abundant in each lake. Adult lake trout abundance was low in Lake Michigan and Parry Sound, and very high at one site in Lake Champlain. Egg deposition was lowest in Lake Michigan (0.4–154.5 eggs•m⁻², median = 1.7), intermediate in Parry Sound (39–1,027 eggs•m⁻², median = 278), and highest in Lake Champlain (0.001–9,623 eggs•m⁻², median = 652). Fry collections in fry traps followed the same trend: no fry in Lake Michigan, 0.005–0.06 fry•trap⁻¹ day⁻¹ in Parry Sound, and 0.08–3.6 fry•trap⁻¹ in Lake Champlain. Egg survival to hatch in overwinter egg bags was similar in Lake Michigan (7.6%) and Parry Sound (2.3–8.9%) in 2001–02, and varied in Lake Champlain (0.4–1.1% in 2001–02, and 1.8–18.2 in 2002–03). Lake trout restoration appears unlikely in northern Lake Michigan at current adult densities, and failure of restoration in Lake Champlain suggests that there are sources of high mortality that occur after fry emergence.

INDEX WORDS: Lake trout, restoration, habitat, spawning, fry emergence.

INTRODUCTION

Populations of lake trout (*Salvelinus namaycush*) were extirpated from most areas of the lower four Great Lakes by the 1960s, and from Lake Champlain (Vermont and New York) by 1900 (Eschmeyer

1957, Plosila and Anderson 1985, Eshenroder 1992, Cornelius *et al.* 1995, Elrod *et al.* 1995, Hansen *et al.* 1995). The causes for the decline in the Great Lakes are generally understood to be commercial fishing and mortality from sea lamprey (*Petromyzon marinus*) attacks; however, the cause of the extirpation of lake trout from Lake Champlain is

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unknown (Ellrott and Marsden 2004). In each of the lakes, restoration of lake trout populations has involved sea lamprey control, annual stocking of one or more strains of lake trout, and regulation of harvest. Lake trout populations became self-sustaining in Lake Superior in the 1990s, but are sustained by stocking in each of the other lakes. Evidence of spawning by stocked fish is extensive in most of the lakes, but there has been little to no evidence of sustained recruitment to age-1 and older except in Lake Superior and at two sites in Lake Huron (Cornelius *et al.* 1995, Elrod *et al.* 1995, Eshenroder *et al.* 1995, Hansen *et al.* 1995, Holey *et al.* 1995, Ellrott and Marsden 2004, Reid *et al.* 2001). Since lake trout stocked as fingerlings or yearlings survive and grow well to maturity, it is generally hypothesized that the impediments to successful lake trout reproduction likely occur between spawning in fall and emergence of fry from spawning reefs in spring.

The purpose of this study was to compare lake trout spawning populations, spawning habitat use, egg deposition, and fry emergence among three lakes representing different stages of lake trout population restoration. In Lake Michigan, the restoration program has been underway since 1965; evidence of reproduction (egg deposition and/or fry hatch) was documented in the Grand Traverse Bay area in the early 1970s, and along the southern shoreline in the early 1980s and 1990s, but year classes of wild age-1+ fish have not been detected (Holey *et al.* 1995). In Lake Huron, restoration stocking began in the early 1970s, and reproduction has been detected since the mid-1980s in South Bay and near Thunder Bay (Anderson and Collins 1995, Johnson and VanAmberg 1995). In Parry Sound, on the north-east shoreline of Georgian Bay, recruitment of naturally-produced fry has been sufficient to produce substantial year classes of wild lake trout since 1988 and restoration was considered to be achieved by 1997 (Reid *et al.* 2001). Lake trout stocking began in Lake Champlain in 1973, and now supports a sport fishery. Starting in 1999, a search for evidence of reproduction by stocked fish revealed numerous sites where spawning and egg survival to hatch occurred, but recruitment of wild fish has been low (Ellrott and Marsden 2004). A comparison among the three lakes provides insight into the size of spawning populations, availability of spawning habitat, use of spawning reefs by stocked fish, and the emergence of fry which are necessary for restoration to succeed. Using Parry Sound as a model of conditions that are adequate

for restoration, we examined progress toward lake trout restoration in lakes Michigan and Champlain by comparing spawning habitat availability, use by spawners, egg densities, and fry hatch. Potential impediments to restoration in Lake Michigan include predation of early life stages (eggs and fry) by exotic species such as alewife (*Alosa pseudoharengus*), round goby (*Neogobius melanostomus*) and rusty crayfish (*Orconectes rusticus*). In Lake Champlain, these exotic predators are absent, but sea lamprey wounding rates are extremely high, such that the amount of energy allocated to spawning may be compromised (Marsden *et al.* 2003).

METHODS

Spawning Site Surveys

In all three lakes, potential spawning sites were located using data on fall aggregations of lake trout, and lake bottom topography and substrate composition. Information on lake trout aggregations was available in historic records from fall commercial fishing and agency assessment work in lakes Michigan and Huron (Goodyear *et al.* 1982, Dawson *et al.* 1997), from modern lake trout assessment work in Lakes Champlain and Huron (Ellrott and Marsden 2004; D. Reid, OMNR, Owen Sound, ON, pers. comm.), and assessment of egg deposition at reefs in Parry Sound (J. Fitzsimons, unpub. data). Spawning reefs used by lake trout generally have deep cobbles that protect eggs over the winter, and a steep slope that may aid in increasing water currents that keep the substrate scoured of silt (Fitzsimons 1994, Marsden *et al.* 1995a). We examined bathymetric charts for areas of steep contour, then examined promising sites using scuba. If rubble/cobble substrates were present, we measured the following characteristics at each site: depth at top and base of the reef, reef area (defined by the area encompassed by cobbles), slope angle of the reef contour, substrate type, particulate substrate size, interstitial depth, presence of silt or organic debris, and presence of zebra mussels. Reef area was measured directly by divers using a 100 m tape, or from GIS maps. Slope angle was measured using a protractor and plumb line. We measured 20 to 60 substrate particles along the longest axis and two perpendicular axes to estimate mean size. In Lake Michigan and Parry Sound each reef was videotaped and substrates were measured from the videotape images. We used the Wentworth scale as described in Marsden *et al.* (1995a) to classify substrate: gravel is < 65 mm, rubble is 65–256 mm,

and cobble is 257–999 mm in the longest dimension. Interstitial depth was measured by digging and inserting a rigid ruler vertically into the substrate several times until a maximum depth was found, or by the depth of the holes that could be created to deploy egg collection bags; interstices likely extend substantially below these depths on most reefs. Substrate shape (angular versus rounded) and presence of silt and zebra mussels were evaluated visually. Silt observed only in interstices was classified as “moderate,” whereas silt covering most or all gravel and rubble particles was classified as “heavy.” Zebra mussel density was low if they covered less than 10% of the substrate, moderate if they covered up to 90% of the substrate in a monolayer, and heavy if there were layers of mussels. We also determined from charts the longest fetch and its direction; this information defines how much water movement can be expected on the reefs due to wind-generated water movements. Water movement is important for keeping reefs free of silt and maintaining dissolved oxygen levels. Subsurface currents were likely present at several reefs, and were highly apparent to divers at the Grand Isle site and offshore locations in northern Lake Michigan, but were not measured. These currents likely vary daily and seasonally; for example, Lake Champlain has a well-documented internal seiche with a 4-day oscillation, in addition to localized seiches (Hunkins *et al.* 1998), and Lake Michigan has a complex seiche with several periods (Mortimer 2004).

Subsequent work in each lake focused on sites that had the highest predicted potential to attract spawning lake trout, based on our assessment of substrate quality, or yielded the largest egg collections in the first year of work. Additional sites were sampled as time allowed. In Lake Champlain the focal sites were Grand Isle and Whallon Bay; in Parry Sound, Davy Is., Bar Is., Horse Is., and Mowat Is.; in Lake Michigan, the Little Traverse Bay crib (LTB crib), Bay Harbor, Fishermens Is., Middle Ground, Dahlia Shoal, and Hog Is. An additional 13 sites in Lake Champlain, one site in Parry Sound, and four sites in Lake Michigan were studied (Fig. 1, Table 1).

Adult Lake Trout Assessment

The density of lake trout aggregated at spawning sites during the spawning period was assessed in fall, 2002, using multipanel white multifilament gillnets set for 1.5 hr between dawn and noon. Each

net consisted of four 30.5 m panels of four mesh sizes (11.4, 12.7, 14, and 15.2 cm stretch mesh), 1.6 m high. Gillnets were fished on the bottom one to three times at two sites in Lake Champlain, five sites in Lake Michigan, and four sites in Lake Huron (Table 2). All lake trout captured were measured (total length), and sex and reproductive status were assessed by expressing gametes. All fish in good condition were released.

Divers made observations of lake trout during the spawning season at all sites where egg bags were buried, with the exception of Parry Sound, and at sites where fall diver reconnaissance was done. These observations provided an evaluation of the presence and relative density of lake trout. Divers made estimates of the number of lake trout seen within an approximate time interval during these dives; while these estimates are not highly quantitative, they do provide an estimate of the differences in density among sites.

Spawning Assessment

To determine whether reefs were used by spawning lake trout, we used two types of egg sampling equipment in the fall of 2000, 2001, and 2002. Egg traps were 20 cm diameter passive egg collectors deployed from the surface in gangs of 30 collectors 1 m apart on a buoyed, anchored line (Horns *et al.* 1989, Marsden *et al.* 1991). These collectors were deployed in late October or early November and checked approximately every 5–10 days. A gang of traps provided non-quantitative data on presence or absence of eggs and relative egg deposition per collector per day (e.g., Schreiner *et al.* 1995). Egg bags were 45.7-cm deep cloth bags constructed of 3-mm Ace mesh attached to a 29.8 cm diameter PVC ring (Perkins and Krueger 1995). The bags were set into a hole dug by a diver into cobble substrate, and were filled with the excavated material. Thirty to 60 egg bags were deployed in early fall before spawning activity began, and retrieved after spawning had mostly or completely ended; the bags therefore provided cumulative, quantitative data on the density of eggs deposited per unit of substrate. The bags were set approximately 1 m apart in lines along the edge of each reef, in areas with the deepest interstitial spaces. Each bag was retrieved by carefully removing substrate to ensure that eggs remained in the bags; the bag was then tied closed with a cable tie before transport to the surface. In 2000 we sampled eggs using traps in Lake Cham-

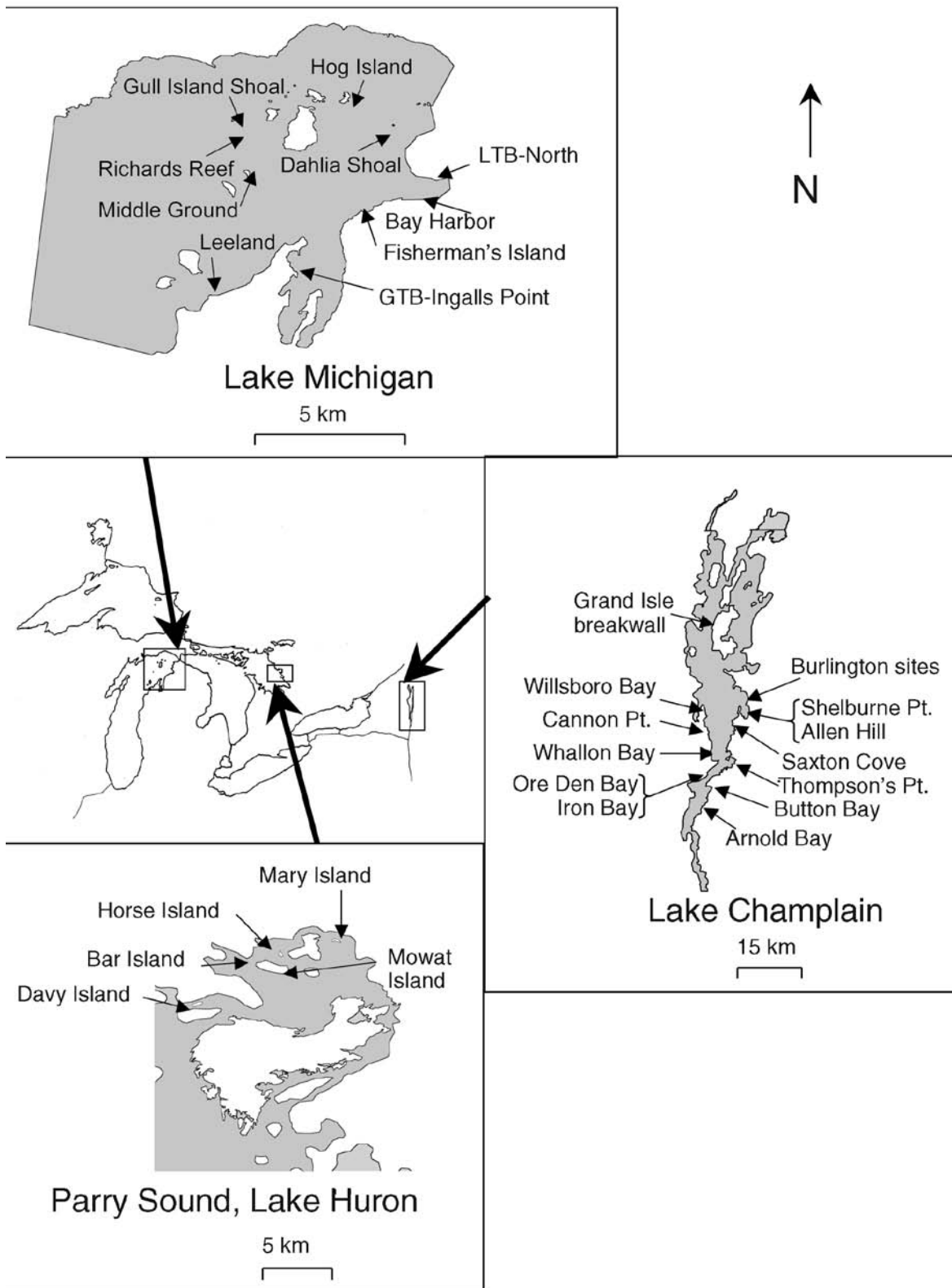


FIG. 1. Map of study sites in Lake Michigan, Parry Sound (Lake Huron), and Lake Champlain. Scales are approximate.

TABLE 1. Summary of habitat characteristics for potential lake trout spawning reefs in Lakes Champlain, Michigan, and Huron (Parry Sound). Site ratings are based on cleanliness of substrate, depth of interstitial spaces, and presence of a steep slope. Area was extrapolated from a combination of actual measurements and visual estimates of reef dimensions. Substrate types were classified by shape (angular or rounded) and particle size (6.5–25.6 cm rubble (R), 25.7–99.9 cm cobble (C), or > 99.9 cm boulder (B)). Zebra mussel density estimates are subjective and classified as high (H), medium (M), and low (L).

Site ^a	Depth range (m)	Approx. Area (m ²)	Slope ^b	Substrate type	Mean substrate size (range) (cm)	Interstitial depth (cm) ^c	Presence of silt ^d	Longest fetch (km) (direction)	Zebra mussel density ^e	Rating
Lake Champlain										
Grand Isle breakwall (A)	0.3–4	53	35–60°	angular R/C	13–99	15–86	low	49 (N)	H	excellent
North of Burlington (A)	0–3.5	23	30–50°	sub-angular C	15–80	>20	low	22 (NW)	L	very good
Burlington breakwall (north) (A)	0.3–8.0	57	30–50°	angular	15–80	>20	low	17 (WNW)	L	very good
Burlington breakwall (south) (A)	0.3–8.0	42	30–50°	angular R/C	15–80	>20	low	21 (WNW)	L	very good
Shelburne Point (N)	0.3–4	—	—	bedrock	—	<10	low	33.5 (NNW)	L	poor
Allen Hill (N)	0.3–8	371	45–65°	angular R/C	15–35	10	high	1 (E)	M	moderate
Willsboro Bay (A)	0.3–18	—	0–45°	angular C/B	40–300	>20	low	22 (NNE)	H	good
Saxton Cove (A)	2.6–5.0	30	10–60°	angular R/C	10–60	10–30	moderate	14.5 (SSW)	H	excellent
Cannon Point (N)	0.3–4.0	—	70°	sub-angular C	30–80	5–15	moderate	66 (N)	M	very good
Whallon Bay (N)	0.3–14	5,980	20–85°	rounded R/C/B	17–106	4–29	low	68 (N)	M	excellent
Thompson's Point (N)	1–10	—	65°	bedrock	—	<10	low	68 (N)	M	poor
Iron Bay (A)	0.3–20	74	0–30°	angular C	10–152	20–100	low	7 (NE)	H	excellent
Ore Den Bay (A)	0.3–20	93	0–10°	angular C	30–90	10–30	low	7 (NE)	H	excellent
Button Bay (N)	1–4	19	—	angular C	—	—	low	17 (SSW)	H	very good
Arnold Bay (A)	1–7	18	60°	angular R/C	10–60	20	moderate	7 (NW)	H	excellent

Site ^a	Depth range (m)	Approx. Area (m ²)	Slope ^b	Substrate type	Mean substrate size (range) (cm)	Interstitial depth (cm) ^c	Presence of silt ^d	Longest fetch (km) (direction)	Zebra mussel density ^e	Rating
Lake Michigan										
Bay Harbor (N)	1–9	11,235	25–50°	rounded R/C	28.3 (<1–390)	30–45	low	104 (NW)	H	very good
Dahlia Shoal (N)	3.5–7	142,010	40–50	rounded R/C	20.3 (0.3–106)	>100	low	82 (SW)	H	excellent
Fishermans Island (N)	1.5–4	195,390	30–45	rounded R/C	24.2 (0.3–131)	>100	low	119 (W)	H	excellent
Gull Island Shoal (N)	1–7	7,600,000	15–30	rounded R/C	5.9 (0.2–6)	>100	low	185 (SW)	H	excellent
Hog Island Reef (N)	2–5	60,493	30–45	rounded R/C	6.5 (0.7–46)	>100	low	104 (SSW)	H	excellent
Ingalls Point (N)	1–1.5	650	50–60	rounded R/C	48.0 (8–293)	20–30	moderate	26 (NE)	H	moderate
Leland (N)	1–2	825	50–60	rounded R/C	28.0 (1.8–96)	20	moderate	96 (W)	H	moderate
LTB Crib (A)	1.5–3.5	100	35–45	rounded R/C	36.4 (2.5–191)	>100	low	22 (SW)	H	excellent
Middle Ground (N)	7–10	245,900	40–50	rounded R/C	45.2 (5–364)	45	low	69 (NW)	H	good
Richards Reef (N)	8–12	1,242,313	20–30	rounded R/C	49.5 (13–112)	>100	low	70 (W)	H	excellent
Parry Sound										
Davy Island (N)	1.5	30	30–40	rounded C	13.9 (4–20)	>30	low	1.1 (SW)	M	excellent
Bar Island (N)	1.5	30	30–40	rounded C	15.1 (4–32)	>30	low	2.4 (SW)	M	excellent
Horse Island (N)	1.6	30	30–40	rounded C	12.3 (5–30)	>30	low	0.7 (SW)	M	excellent
Mowat Island (N)	1.8	30	25–30	rounded C	13.0 (4–25)	>30	low	3.5 (SW)	M	moderate
Mary Island (N)	0.9	30	20–30	rounded C	13.8 (4–28)	>30	low	6.5 (SW)	M	moderate

^a N = natural reef, A = artificial (man-made) reef.

^b A slope of 0° is vertical; 90° is horizontal.

^c Measured vertically using a rigid ruler; interstices may extend below these depths.

^d Silt was classified as low if it was barely visible, moderate if there was a thin coating of silt on hard substrates, and high if there was > 1 cm coating of silt on hard substrates.

^e Zebra mussel density was classified as low if mussels were scattered on the substrate, moderate if there was a monolayer of mussels, and high if there were dense, multi-layer colonies that occluded interstitial spaces.

TABLE 2. Summary of fall 2002 gillnet sampling for adult lake trout in lakes Champlain, Huron, and Michigan. All nets were set between 0730 and 1200 EST. M = male, F = female, U = unknown sex; ND = no data (fish fell out of net before capture). Species codes LAW = lake whitefish, SMB = smallmouth bass, SL = sea lamprey.

Lake/Site	Date	Depth (m)	Number of lake trout (TL-mm, sex)	Other species
Lake Champlain				
Grand Isle	12 Nov	3–11.6	1 (668M)	5 LAW
	12 Nov	3–7	4 (720F, 580M, 697M, 605U)	2 LAW, 2SL
Whallon Bay	13 Nov	3–5.2	3 (752U, 673M, 592M)	1 LAW
	14 Nov	1.5–36.9	0	2 SMB
	14 Nov	2–18	2 (655F, ND)	4 SMB
Total effort = 7.5 h				
Total CPUE = 1.3 lake trout•h				
Lake Michigan				
Fisherman's Island	17 Oct	6–8	0	0
	21 Oct	10–16	0	0
Gull Island Reef	24 Oct	15–18	0	0
	25 Oct	21–23	0	0
	05 Nov	20–22	0	0
Richards Reef	24 Oct	30–38	0	0
	25 Oct	26–30	0	0
	05 Nov	24–32	0	0
Bay Harbor	17 Oct	8–16	0	0
	18 Oct	10–16	0	0
LTB Crib	17 Oct	12–18	0	0
	18 Oct	15–19	0	0
	31 Oct	14–15	0	0
Total effort = 19.1 h				
Total CPUE = 0 lake trout•h				
Parry Sound				
Davy Island	22 Nov	2–18	0	0
	23 Nov	2–18	0	0
	24 Nov	2–18	0	0
Mowat Island	22 Nov	4–13	0	334
	23 Nov	2–25	0	0
	24 Nov	2–23	1 (681M)	0
Bar Island	22 Nov	2–10	0	0
	23 Nov	2–10	1 (661F)	0
	24 Nov	2–11	0	0
Horse Island	22 Nov	2–21	1 (572U)	316(3)
	23 Nov	2–22	0	91, 334
	24 Nov	2–22	0	0
Total effort = 18 h				
Total CPUE = 0.17 lake trout•h				

plain, and egg bags in Lake Michigan; we used egg bags in all three lakes in 2001 and 2002.

Microhabitat Selection

Egg collection bags were generally set along the edge of each reef, at the top of a slope. To test the assumption that this is the area where lake trout focus spawning effort, we also set 15 egg bags across the contour at Whallon Bay, Lake Champlain, from 2.3 to 10 m depth, in 2002. This line of bags crossed the center of the line of bags set along the contour.

Egg Survival and Fry Hatch

Survival of eggs over the winter was assessed using fine-mesh (1.5 mm) egg bags that would retain hatched fry. Thirty egg bags were deployed at each site in early fall as described above. The bags were closed with a 1.5 mm mesh, close-fitting cover at the end of the spawning season, and retrieved in spring as soon as ice thaw allowed access to the reefs (mid to late April). The bags were retrieved by leaving the cover on and substrate inside to ensure that hatched fry did not escape. Presence of any holes in the bags or covers that may have allowed escape of fry was noted as they were processed; if large holes were present, the bag contents were not included in the analysis. Overwinter egg bags were used at two to four sites in each lake in 2001–02 and at two sites in Lake Champlain in 2002–03.

Fry densities on the reefs were also assessed in spring using two designs of emergent fry traps. Both designs consisted of a pyramid of metal mesh, open at the bottom, with a capture bottle at the top. The capture bottle had an inverted funnel set into the opening, so that fry which emerged from substrate beneath the trap and entered the bottle would be unlikely to escape. Emergent fry traps described by Collins (1975), and modified by Marsden *et al.* (1988), were deployed with buoyed lines from the surface, and were checked by lifting them to the surface from a boat. A second design constructed of perforated stainless steel had a 12 cm square channel around the outside edge, into which rocks were placed to help maintain the trap in place. These traps remained in place and were checked by divers, who would remove and replace the capture bottles and check each recovered bottle for fry at the surface. Both types of traps were deployed as early in the season as ice thaw allowed, and were

checked for fry every 7–16 days until no more fry were captured. Diver traps were set approximately 1 m apart in lines along the reef contour, in a similar position to the egg bags; we attempted to place the surface fry traps in the same location, but there was some variation in their location from set to set. Fry traps were used at two to four sites in each lake, and were deployed in Lakes Champlain and Michigan in 2001, and in all three lakes in 2002 and 2003.

Data Analysis

Statistical comparisons of physical traits of sites within and among lakes were not made due to non-random selection of sites and extremely high variation, which obscured differences among lakes. Selection of sites within lakes and location of collecting equipment at each site were targeted at optimal spawning substrate and contour to focus efforts in areas where we had the highest probability of acquiring spawner density, egg and fry data.

RESULTS

Spawning Site Surveys

In Parry Sound and Lake Michigan, all sites surveyed had rounded substrates, except for Bar Island where substrates were more angular. While there was a wide range of particle sizes, particularly in Lake Champlain, substrates were mostly rubble and small cobbles. In Lake Michigan substrate was primarily limestone, whereas in Parry Sound it was granite. In Lake Champlain the surficial geology is more diverse and is comprised of marble, conglomerate, quartzite, granite, and dolomite. Natural sites were characterized by rounded and angular cobble/rubble substrate, or flat bedrock (Table 1). Silt and organic material were absent or present in small quantities; over all sites, Lake Michigan had more zebra mussels, periphyton, and siltation than the other two lakes (Table 1). In Lake Champlain, fine silts were not present in shallow water (< 22 m) except at Allen Hill (0.3–0.8 m), which had the shortest fetch and was thickly coated with fines. At sites where surveys extended below 22 m (Willsboro Bay, Ore Den Bay, Iron Bay) we found thick (> 2 cm) deposits of silt that filled interstitial spaces. Ten of the fifteen sites in Lake Champlain and one site surveyed in Lake Michigan were man-made (breakwalls, mine tailings, water intake lines covered with rip-rap, and a ruined pier), with angular rubble-cobble substrates (Table 1). Man-made

sites were in shallow water (< 7 m). Of the sites where fall aggregations of lake trout had been reported, only two were classified as poor spawning sites (Shelburne Point and Thompson's Point); both were in Lake Champlain, and were primarily bedrock with no interstices.

Adult Lake Trout Assessment

Gillnet surveys captured 1.3 lake trout•hr⁻¹ over two reefs in Lake Champlain and 0.2 lake trout•hr⁻¹ over four reefs in Parry Sound (Table 2). All lake trout were sufficiently large to be mature (572 to 752 mm TL) and both sexes were collected in both lakes. No lake trout were caught during 19.1 hr of gillnetting over five reefs in Lake Michigan, an effort greater than either of the other two lakes. Other species caught in gillnets included lake whitefish (*Coregonus clupeaformis*) and smallmouth bass (*Micropterus dolomieu*); both are species which have seasonal preferences for rocky substrates, similar to lake trout.

Observations of adult lake trout aggregating at spawning sites provided an alternate method for assessing relative lake trout abundance (e.g., Marsden and Janssen 1997). In Lakes Michigan, Huron, and Ontario, lake trout were observed in the range of 0.2 to 2 lake trout seen per minute during the day (Marsden and Janssen 1997). In our investigation, no lake trout were observed in Parry Sound, and low numbers were seen at two sites (LTB crib and Bay Harbor) in Lake Michigan. A maximum of approximately 20 fish were seen in one 1-hr dive at the LTB crib, and five fish at Bay Harbor. In Lake Champlain, lake trout were occasionally observed at Whallon Bay while diving, and at Arnold Bay from the surface (less than two per hour). Lake trout were both seen and heard leaping at the surface during the evening at Whallon Bay, Grand Isle, and Arnold Bay. At Grand Isle, large numbers of lake trout were regularly seen along the distal third of the breakwall in dense, horizontal columns of fish approximately 5–7 fish wide extending over 10 m. Divers estimated that several hundred fish were present at the site; there were too many to count with confidence. These fish were not sampled in our daytime gillnet sets, likely because the nets were highly visible in shallow water during the day.

Spawning Assessment

Eggs were collected at all five sites sampled in Parry Sound, nine of 15 sites sampled in Lake

Champlain, and five of ten sites in Lake Michigan. Reefs where we found evidence of spawning had cobble/rubble substrate with interstitial spaces at least 20 cm deep, with a steep slope and absence of fines (Table 1). A single egg was collected on one of the two sites considered to be poor quality (Shelburne Bay in Lake Champlain).

Catch per unit effort (CPUE) and numbers of eggs collected per square meter varied substantially among reefs within each lake, among years at each reef, and also among lakes (Table 3). Across all sampling years, egg densities were the lowest in Lake Michigan (0.4–154.5 eggs•m⁻²), intermediate in Parry Sound (39–1,027 eggs•m⁻²), and highest overall in Lake Champlain but with higher variation among sites (0.001–9,623 eggs•m⁻²; Table 3).

Use of egg traps and nets in Lake Champlain provided data on when eggs were being deposited (Table 3). The last date on which egg traps and nets were retrieved with eggs in them was between late November and mid-December. Egg bags were retrieved each year between mid-November and early December; thus, we may not have collected eggs deposited late in the spawning season. This would reduce the total count of eggs, but likely does not affect the relative estimates of egg deposition among reefs.

Microhabitat Selection

Eggs were not always deposited uniformly, even on apparently homogeneous substrate. The line of bags set across the contour at Whallon Bay in Lake Champlain collected 709 eggs; the bags at either end of the line had few (< 15 per bag) or no eggs, whereas the bags in the center of the line, at the top of the reef slope, contained the most eggs (45 to 243 eggs per bag). In contrast, there was no obvious pattern of egg abundance in bags in the very homogeneous sites at Parry Sound, although there was considerable variation among bags (Table 3).

Egg Survival and Fry Hatch

Fry that had overwintered in egg bags were collected on both reefs sampled in Lake Champlain, two of three sampled in Parry Sound, and one of four sites sampled in Lake Michigan. Fry collections in egg bags were highest at Grand Isle (mean 107 fry•m⁻²) in Lake Champlain, and similar among the LTB crib (12 ± 26 fry•m⁻²) in Lake Michigan and Davy (12 ± 27 fry•m⁻²) and Bar (12 ± 32 fry•m⁻²) Island sites in Parry Sound. Fry

TABLE 3. Summary of lake trout egg deposition in Lake Champlain, Parry Sound, and Lake Michigan, 2000–02. Eggs were collected in egg traps or in egg bags buried in the substrate. Collections in egg traps are reported as mean number of eggs•trap⁻¹•day⁻¹; collections from egg bags are reported as mean number of eggs•m⁻² ± SD.

Year and location	Gear type	No. traps/bags	Collection dates ^a	Number of eggs	Number•m ⁻² or trap ⁻¹ •day ⁻¹
2000					
Lake Champlain					
Grand Isle	trap	30	13 Nov–8 Dec	3,799	4.08 ^b
Cannon Point	trap	30	14 Nov–29 Nov	0	0
Whallon Bay	trap	30	14 Nov–28 Nov	1	0.002 ± 0.03
Button Bay	trap	30	19 Nov–11 Dec	10	0.02 ± 0.21
Arnold Bay	trap	30	3 Nov–11 Dec	85	0.12 ± 0.33
Lake Michigan					
Bay Harbor	bags	59	29 Nov	4	0.95 ± 3.55
Dahlia Shoal	bags	59	29 Nov	0	0
Fisherman's Island	bags	46	29 Nov	0	0
Hog Island Reef	bags	60	2 Dec	0	0
Ingalls Point	bags	30	27 Nov	2	0.93 ± 3.55
LTB Crib	bags	33	27 Nov	362	154.48 ± 255.97
Middle Ground	bags	60	14 Dec	0	0
2001					
Lake Champlain					
Grand Isle	bags	59	27 Nov	39,593	9,623 ± 1,658
Burlington Breakwall (N)	bags	15	3 Dec	0	0
Burlington Breakwall (S)	bags	15	29 Nov	0	0
Shelburne Point	traps	30	30 Oct–29 Nov	1	0.001 ± 0.03
Willsboro Bay	bags	19	5 Dec	0	0
Allen Hill	bags	15	29 Nov	0	0
Saxton Cove	bags	15	30 Nov	116	118 ± 59
Cannon Point	bags	15	4 Dec	0	0
Whallon Bay	bags	60	4 Dec	2,728	652 ± 158
Thompson's Point	traps	30	8 Nov–13 Nov	0	0
Iron Bay	bags	15	4 Dec	2	1.9 ± 1.3
Arnold Bay	bags	15	30 Nov	949	972 ± 289
Lake Michigan					
Bay Harbor-deep	bags	58	9 Nov	0	0
Bay Harbor-shallow	bags	37	13 Nov	2	0.76 ± 4.60
Bay Harbor	bags	37	21 Nov	16	6.05 ± 14.21
Dahlia Reef	bags	38	29 Nov	0	0
Fisherman's Island	bags	85	26 Nov	35	5.78 ± 15.83
Gull Island Shoal	bags	46	11 Dec	6	1.83 ± 5.61
Hog Island	bags	42	29 Nov	0	0
Ingalls Point	Bags	70	29 Nov	2	0.40 ± 3.35
LTB crib	bags	30	19 Nov	107	50.17 ± 53.85
Parry Sound					
Davy Island	bags	29	14 Nov	377	186 ± 224
Bar Island	bags	29	14 Nov	775	519 ± 411
Horse Island	bags	29	14 Nov	563	278 ± 258
Mowat Island	bags	29	14 Nov	577	284 ± 362
Mary Island	bags	24	14 Nov	67	39 ± 46

(Continued)

TABLE 3. (Continued).

Year and location	Gear type	No. traps/bags	Collection dates ^a	Number of eggs	Number•m ⁻² or trap ⁻¹ •day ⁻¹
2002					
Lake Champlain					
Grand Isle	bags	60	26 Nov	16,037	3,832 ± 3,571.1
Whallon Bay	bags	45	25 Nov	1,033	342.8 ± 872.5
Lake Michigan					
Bay Harbor	bags	105	5, 13, 18 Nov	30	4.02 ± 16.45
Dahlia	bags	58	26 Nov	0	0
Fisherman's Island	bags	51	20 Nov	8	2.18 ± 8.10
Gull Island Shoal-east	bags	57	26 Nov	0	0
Gull Island Shoal-west	bags	54	26 Nov	3	0.78 ± 4.23
Hog Island	bags	59	26 Nov	0	0
Ingalls Point	bags	60	19 Nov	0	0
Leeland	bags	48	19 Nov	0	0
LTB crib	bags	60	18 Nov	137	32.17 ± 65.94
Middle Ground	bags	57	26 Nov	0	0
Richards Reef	bags	59	26 Nov	0	0
Parry Sound					
Davy Island	bags	60	13 Nov	4,297	1,027 ± 1,052
Horse Island	bags	59	13 Nov	776	189 ± 228.6
Mowat Island	bags	60	13 Nov	783	187 ± 271.5
Bar Island	bags	60	13 Nov	3,975	950 ± 990.8

^aEgg bags were all deployed in late summer prior to the onset of spawning; only retrieval dates are given. Egg traps were usually deployed prior to egg deposition; only the range of dates when eggs were collected are listed.

^bEggs from traps were not tallied separately on this date, so the SE could not be calculated.

emergence as measured by fry traps yielded similar results – high fry density and CPUE in Lake Champlain (0 to 907.5 fry• m⁻² and 0 to 3.6 fry• trap⁻¹• day⁻¹), lower densities in Parry Sound (0 to 9• m⁻² and 0 to 0.06 fry• trap⁻¹• day⁻¹), and no emergent fry captured in Lake Michigan (Table 4). The number of fry emerging per 100 eggs estimated from fry traps was 0.12 to 6.8 in Parry Sound in 2002 and 2003, and 11.7 to 14.5 in Lake Champlain in 2003. Estimates of fry hatch (fry per 100 eggs) measured with egg bag collections and emergence measured with fry traps followed similar patterns at most sites: 6.8 and 8.9, respectively, at Davy Is. and 0.8 and 2.3 at Bar Is. in Parry Sound in 2001–02, 14.5 and 18.2 at Grand Is. in Lake Champlain in 2002–03 (Tables 4 and 5). The number of hatched fry in egg bags compared with egg density estimated in bags removed in fall was lowest in Lake Champlain (0.4–1.1%), and similar in Lake Michigan (7.6%) and Parry Sound (2.3–8.9%) in 2001–02 (Table 5).

DISCUSSION

Overall, we found that the proportion of reefs visited by lake trout, number of adult lake trout, density of eggs deposited, and number of fry hatched were highest in Lake Champlain, intermediate in Lake Huron, and the lowest in Lake Michigan. The availability and quality of spawning sites appear to be similar in all three lakes, but measured egg densities vary substantially among lakes and among sites within lakes. While these results explain the absence of detectable recruitment in Lake Michigan, it is more difficult to explain why recruitment is low in Lake Champlain compared with Parry Sound.

Potential spawning sites meeting criteria for lake trout spawning based on the literature and the authors' unpublished observations were readily found in all three lakes (Fitzsimons 1994, 1995, 1996; Gunn *et al.* 1996; Marsden 1994; Marsden *et al.* 1991, 1995a and b). The total reef area available

TABLE 4. Lake trout fry collections in fry traps in Lake Champlain and Parry Sound, 2001–03. Fry were collected in three designs of emergent fry traps: steel traps checked by divers (diver), and steel traps or soft traps checked by lifting to the surface (surface). Fry•m⁻² was calculated as total number of fry divided by the total area under all the traps at a given site. Number of fry collected per trap-day was calculated only over the range of dates when fry were collected in traps. Number of fry•100 eggs⁻¹ was calculated from quantitative estimates of eggs (from fall egg bags—Table 1) divided into quantitative estimates of fry (from diver traps) × 100. Surface fry traps did not yield quantitative estimates of fry per unit area. Na = data not available.

Year/ Site	Trap type	Date deployed	Number of traps	Number of fry	Fry•m ⁻²	Fry•trap ⁻¹ •day ⁻¹	Eggs•m ⁻²	Fry•100 eggs ⁻¹
2001								
Lake Champlain								
Grand Isle	surface	16 Apr–16 Jun	10	173	66.5	0.33	na	na
Whallon Bay	surface	20 Apr–13 Jun	19	148	30	0.14	na	na
Ore Den Bay	surface	3 May–19 Jun	10	0	0	0	na	na
Arnold Bay	surface	2 May–4 Jun	10	594	228.5	2.38	na	na
Lake Michigan								
LTB crib	diver	05 Apr–5 May	16	0	0	0	0	0
Bay Harbor	diver	05 Apr–5 May	24	0	0	0	0	0
2002								
Lake Champlain								
Grand Isle	surface	05 Apr–6 Jun	12	237	76	0.32	na	na
Saxton Cove	surface	26 Apr–3 Jun	7	0	0	0	na	na
Whallon Bay	surface	10 Apr–4 Jun	20	89	17.1	0.08	na	na
Arnold Bay	surface	26 Apr–28 May	6	249	159.6	1.73	na	na
Lake Michigan								
LTB crib	diver	10 Apr–10 Jun	16	0	0	0	0	0
Bay Harbor	diver	10 Apr–10 Jun	24	0	0	0	0	0
Fisherman's Island	diver	10 Apr–10 Jun	24	0	0	0	0	0
Parry Sound								
Davy Island	diver	16 Apr–18 Jun	8	25	9.3	0.05	136	6.8
Bar Island	diver	16 Apr–18 Jun	8	11	4.1	0.02	519	0.8
Horse Island	diver	16 Apr–18 Jun	8	3	1.1	0.01	277	0.4
Mowat Island	diver	16 Apr–18 Jun	8	0	0	0	219	0
2003								
Lake Champlain								
Grand Isle	diver	22 Apr–12 Jun	12	463	907.5	1.10	3,183	14.5
Whallon Bay	diver	22 Apr–19 Jun	13	152	297.9	0.29	1,299	11.7
Grand Isle	surface	23 Apr–12 Jun	7	176	96.7	0.74	na	na
Saxton Cove	surface	30 Apr–4 Jun	10	4	1.5	0.02	na	na
Arnold Bay	surface	29 Apr–10 Jun	10	1,008	387.7	3.60	na	na
Parry Sound								
Davy Island	diver	25 Apr–11 Jun	14	37	8.0	0.06	1,017	0.79
Bar Island	diver	25 Apr–11 Jun	14	5	1.2	0.01	941	0.12
Horse Island	diver	25 Apr–11 Jun	14	3	0.6	0.01	187	0.32
Mowat Island	diver	25 Apr–11 Jun	14	5	1.1	0.01	171	0.63
Lake Michigan								
Fisherman's Island	diver	17 Apr–15 May	12	0	0	0	0	0

TABLE 5. Lake trout fry collections in egg bags retrieved in spring from Lake Champlain and Parry Sound, 2001–02 and 2002–03. No egg bags were left overwinter in Lake Michigan. Mean percent hatch (number of fry per 100 eggs) was calculated from quantitative estimates of eggs (from fall egg bags—Table 1) divided into the mean fry/m² × 100.

Site	# bags	Dates deployed	Mean # Fry	Mean fry/m ²	% hatch
2001–02					
Lake Champlain					
Grand Isle	29	10 Oct–1 Apr	3,006	107 ± 98	1.1
Whallon Bay	26	21 Aug, 6 Sept–11 Apr	72	2.8 ± 8.0	0.4
Lake Michigan					
LTB Crib	30	20 Sept–16 Apr	22	12 ± 25.6	7.6
Bay Harbor	24	20 Sept–11 Apr	0	0	0
Parry Sound					
Davy Island	25	4–5 Sept–16 Apr	23	12 ± 26.9	8.9
Bar Island	27	4–5 Sept–16 Apr	22	12 ± 32.0	2.3
Mary Island	14	4–5 Sept–16 Apr	0	0	0
2002–03					
Lake Champlain					
Grand Isle	29	27 Aug–25 Apr	1,013	581 ± 85	18.2
Whallon Bay	27	9 Sept–22 Apr	44	23 ± 12.6	1.8

for lake trout spawning appears to be highest in Lake Michigan, intermediate in Lake Champlain, and smallest in Parry Sound. In general, sites in Lake Michigan had higher levels of periphyton and higher water clarity than in Lake Champlain; water clarity in Parry Sound was greater and the substrate was cleaner than in the other two lakes. Generally, there were no obvious differences in substrate size, cleanliness, reef contour, or reef size between sites with evidence of spawning, and sites without. High zebra mussel densities and high turbidity can negatively affect egg incubation by reducing entrapment of eggs (Marsden and Chotkowski 2001) and increasing siltation; however, sites in Lake Champlain with dense zebra mussels and low water clarity produced some of the highest fry hatch per egg seen in this study. More artificial sites than natural sites (6 versus 3) were used by spawning lake trout in Lake Champlain, and the only artificial site investigated in Lake Michigan produced the highest egg densities (LTB crib). No artificial sites were sampled in Parry Sound. The highest densities of eggs and fry were recovered from the smallest sites in lakes Champlain and Michigan. Sites in Parry Sound were of similar size and small relative to other lakes (Tables 1, 3). It is possible that smaller sites concentrated spawning activity and eggs.

Catches of adult lake trout in gillnets, while small, confirmed the patterns observed for egg deposition; the highest numbers were captured in Lake Champlain and no lake trout were captured in Lake Michigan surveys. There was a higher likelihood of seeing lake trout at smaller sites (Grand Isle, Arnold Bay, LTB crib), an indication of concentration around limited habitat. The absence of lake trout observed in Parry Sound is particularly puzzling, as the sites were very limited in area, and divers traversed the entire area of available spawning substrate during a related project during the spawning season.

Our ability to identify lake trout spawning sites, as measured by the proportion of sites that yielded eggs and fry, was high. Given the intensive sampling effort over three years with several types of gear, focused best quality habitat the absence of eggs or fry likely signified low spawning activity. The distribution of egg collections in the cross-contour line of egg bags at Whallon Bay, Lake Champlain, confirms Perkins and Krueger's (1995) observation that eggs tend to be deposited at the edge of the downward slope. We generally set egg bags in the optimum location at each site, given that we focused on sites with a steep slope.

The extreme spawning density noted at the Grand Isle site in Lake Champlain likely results from at-

traction to effluent from a salmonid hatchery which enters the lake approximately 200 m from the breakwall (Ellrott and Marsden 2004). The highest egg density in a single egg bag was $31,577 \cdot \text{m}^{-2}$; in 2001 we observed windrows of an estimated several million eggs along the base of the breakwall at 7 m deep. These eggs were probably deposited on the breakwall and drifted to the base. While the high densities of zebra mussels on the breakwall may have inhibited entrainment of eggs into interstices (Marsden and Chotkowski 2001), the extraordinary number of loose eggs may also indicate saturation of the interstitial spaces on the reef. Two weeks after the windrows were observed all eggs at the base of the breakwall were gone, presumably moved by currents and/or eaten by predators. Large schools of yellow perch (*Perca flavescens*) foraging intensively on the breakwall substrate, were videotaped at this site in late November, 2004; while we did not sample these fish, Fitzsimons (1990) documented yellow perch eating large quantities of lake trout eggs in Keuka Lake.

There was substantial variation in egg densities within lakes and among years. Bags were buried in nearly identical locations in Parry Sound during 2001 and 2002 and substrate within all of the sites at Parry Sound was very homogeneous. Despite consistencies in sampling methodology, egg collections increased nearly 5-fold at Davy Is. (from a mean of 186 to $1,027 \text{ eggs} \cdot \text{m}^{-2}$) from 2001 to 2002, and decreased at Horse Is. and Mowat Is. Similarly, at Grand Isle in Lake Champlain egg collections decreased from a mean of $9,623 \text{ eggs} \cdot \text{m}^{-2}$ in 2001 to $3,183 \text{ eggs} \cdot \text{m}^{-2}$ in 2002. This within- and between-year variation in reproductive output may not affect long-term sustainability of lake trout populations in lakes where the mean egg density is high, but should be considered in annual estimates of recruitment potential.

Egg density and egg survival from Lake Champlain and Parry Sound were comparable to or greater than those measured in the other Great Lakes, whereas egg densities in Lake Michigan were substantially lower. Mean egg density in egg bags or similar samplers has been measured at $122\text{--}518 \text{ eggs} \cdot \text{m}^{-2}$ in Lake Superior (Peck 1986), $6\text{--}6,178 \text{ eggs} \cdot \text{m}^{-2}$ in Lake Ontario (Fitzsimons 1995, Perkins and Krueger 1995), and $80\text{--}12,605 \cdot \text{m}^{-2}$ in southern Lake Michigan (Marsden and Chotkowski 2001); the range we measured was $118\text{--}9,623 \text{ eggs} \cdot \text{m}^{-2}$ (median = 652) in Lake Champlain, $39\text{--}1,027 \text{ eggs} \cdot \text{m}^{-2}$ (median = 278) in Parry Sound, and $0.4\text{--}154.5 \cdot \text{m}^{-2}$ (median = 1.7) in

northern Lake Michigan. These data suggest that egg density alone is not a good predictor of recruitment; for example, egg density in Parry Sound was lower than in southern Lake Michigan, where no recruitment has been seen.

Fry hatch and emergence, measured with egg bags and fry traps respectively, were highest in Lake Champlain, intermediate in Parry Sound, and lowest in Lake Michigan. Numbers of fry collected in egg bags left in the substrate over winter reflect loss of eggs due to fungus and other sources of mortality. Fry traps provide relative measures of abundance between hatching and emergence. If we assume that fry catches in traps which remain in place throughout the emergence period, represent only the number of fry emerging from a limited area under and adjacent to each trap, it is possible to compare with fall egg densities and estimate a relative proportion of eggs that hatched. This comparison indicates that substantially more fry emerged per 100 eggs in Lake Champlain than Parry Sound (Table 4). However, fry move laterally within the substrate prior to emergence (Marsden, unpublished data), variations in habitat structure may bias this assumption. To account for variation, we also determined fry hatch from egg bags that were covered and left in the substrate overwinter. These data suggest a low fry hatch in Lake Champlain relative to the other two lakes. Hatch rates were likely underestimated at Grand Isle, Lake Champlain, in 2001–02 as large numbers of eggs deposited in the bags formed masses of fungus resulting in near complete mortality. In 2002–03, egg deposition was lower at Grand Isle, and an estimated 18.2% of the eggs hatched. Similarly, fry traps at Whallon Bay collected a considerably higher numbers of fry (11.7 fry per 100 eggs) relative to egg bags (1.8 fry per 100 eggs); this difference may be attributable movement through the substrate or high mortality of eggs in the egg bags.

At Arnold Bay, Lake Champlain, swarms of sac fry were observed by divers when the substrate was disturbed; given that lake trout fry are rarely ever seen by divers (Marsden and Fitzsimons, pers. obs.), this observation suggests that fry hatch was extremely high at this site. Estimates of egg survival in Lake Champlain were similar to measurements in Lake Ontario, which ranged from 12.4–16.5% (Perkins and Krueger 1995). Densities of fry measured with emergent fry traps in Lake Ontario ($2\text{--}289 \text{ fry} \cdot \text{m}^{-2}$) and Lake Superior ($20\text{--}46 \text{ fry} \cdot \text{m}^{-2}$) were generally higher than those we mea-

sured in Parry Sound ($0.6\text{--}9.3 \text{ fry}\cdot\text{m}^{-2}$) but similar to or lower than densities in Lake Champlain ($1.5\text{--}907.5 \text{ fry}\cdot\text{m}^{-2}$; Fitzsimons *et al.* 2004).

If we use Parry Sound as a benchmark for successful restoration of trout, the low use of spawning reefs in northern Lake Michigan indicates that the density of spawning lake trout is too low to support natural recruitment (Bronte *et al.* 2003a, 2003b). Lake trout assessments suggest that adult stocks are too low in northern Lake Michigan for restoration to occur (Bronte *et al.* 2003a). Selgeby *et al.* (1995) conducted a comprehensive synthesis of gillnet catch-rates to define a relative index of spawning lake trout necessary to produce measurable recruitment ($17 \text{ to } 135 \text{ lake trout}\cdot 305 \text{ m}^{-1}$ of gillnet). Adult abundance levels measured at historically important offshore spawning locations in northern Lake Michigan in 1999–2001 ranged from 0 to 145, but most sites were well below acceptable levels for recruitment and the majority of sites were not even visited by spawning lake trout (Bronte *et al.* 2003a). Low stocking rates, high mortality, or combinations of both likely contribute to the low abundance of adult lake trout in northern Lake Michigan. Exploitation by sport and tribal fisheries had been high, and in recent years mortality due to sea lamprey has increased substantially.

In contrast to Lake Michigan, spawning reef use, abundance of adults, egg density, and fry hatch are high enough, in Lake Champlain than would be needed to support recruitment. Data on adult abundance to compare with the Great Lakes fall gillnetting data are not available, as fall assessments use electroshocking and gillnet surveys were only conducted in summer. The proportion of unclipped lake trout in the adult population in Lake Champlain is low ($< 4\%$), with the apparent exception of one year class (Ellrott and Marsden 2004). Previous work on lake trout restoration has focused on bottlenecks to survival during the stages between egg deposition and emergence. Data from Lake Champlain suggest that high mortality is likely occurring after fry emerge. Further research in Lake Champlain and other lakes that also lack exotic species as a source of post-emergent mortality may further our understanding of mechanisms driving lake trout recruitment failures in the Great Lakes.

The abundance of adult lake trout necessary for restoration may be related to the amount and size of spawning habitat available. At one extreme, high densities of eggs over limited habitat may be detri-

mental to egg survival, due to transmission of fungus among densely packed eggs, and saturation of interstices (Ellrott and Marsden 2004). In contrast, in northern Lake Michigan, a similar abundance of spawners over the many large spawning reefs may produce egg densities that could sustain reproduction (Jonas *et al.* in press). To achieve measurable fry production, stocking efforts need to focus on achieving optimal spawner densities per unit area of habitat. Research efforts should focus on investigating factors, that affect site fidelity and attraction to spawning reefs. In large lakes, target spawner abundance levels should be tailored to match regional and reef specific characteristics rather than striving to rebuild lake-wide populations.

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